

Yifan Zhou

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EDUCATION

University of Georgia

Doctor of Philosophy in Computer Science

Aug 2024 – Aug 2028

Johns Hopkins University

Master of Science in Electrical & Computer Engineering

Aug 2022 - Dec 2023

University of North Carolina at Chapel Hill

Bachelor of Science in Computer Science

Aug 2020 - May 2022

PUBLICATION

- Xue, X., **Zhou, Y.**, Wang, Z., Tang, S., Torr, P., Ouyang, W., Bai, L., & Yin, Z. (2026). StraTA: Incentivizing Agentic Reinforcement Learning with Strategic Trajectory Abstraction. arXiv preprint arXiv:2605.06642 (under submission).
- Xue, X., **Zhou, Y.**, Zhang, G., Zhang, Z., Li, Y., Zhang, C., Yin, Z., Torr, P., Ouyang, W., & Bai, L. CoMAS: Co-Evolving Multi-Agent Systems via Interaction Rewards. In Proceedings of the International Conference on Learning Representations (ICLR 2026).
- Tan, Z., Geng, H., Yu, X., Zhang, M., Wan, G., **Zhou, Y.**, He, Q., Xue, X., Zhou, H., Fan, Y., Li, Z.-Z., Zhang, Z., Zhang, G., Zhang, C., Yin, Z., Torr, P., & Bai, L. (2026). Scaling Behaviors of LLM Reinforcement Learning Post-Training: An Empirical Study in Mathematical Reasoning. In Proceedings of the Annual Meeting of the Association for Computational Linguistics (ACL 2026 Main).
- Zhang, G., Geng, H., Yu, X., Yin, Z., Zhang, Z., Tan, Z., Zhou, H., Li, Z.-Z., Xue, X., Li, Y., **Zhou, Y.**, Chen, Y., Zhang, C., Fan, Y., Wang, Z., Huang, S., Piedrahita Velez, F., Liao, Y., Wang, H., Yang, M., et al. (2026). The Landscape of Agentic Reinforcement Learning for LLMs: A Survey. Transactions on Machine Learning Research (TMLR).
- Sun, B., **Zhou, Y.**, & Jiang, H. (2025). Empowering Users in Digital Privacy Management through Interactive LLM-Based Agents. In Proceedings of the International Conference on Learning Representations (ICLR 2025).
- Pan, Y., Jiang, H., Chen, J., Li, Y., Zhao, H., **Zhou, Y.**, Shu, P., Wu, Z., Liu, Z., Zhu, D., Li, X., Abate, Y., & Liu, T. (2025). EG-SpikeFormer: Eye-Gaze Guided Transformer on Spiking Neural Networks for Medical Image Analysis. In Proceedings of the IEEE International Symposium on Biomedical Imaging (ISBI 2025). (Oral)

RESEARCH EXPERIENCE

Research Intern | MGH, Harvard Medical School, CAMCA Lab

May 2025 - Now

(Agentic RL, Large Language Model)

- Applied reinforcement learning techniques to train LLM-based agents for OCD-related intervention tasks, supporting personalized and structured patient guidance.
- Conducted research on agentic reinforcement learning (Agentic RL) for large language models in mental health applications.
- Worked with mainstream RL training frameworks including RLLM and VeRL and became familiar with the end-to-end RLHF / RL for LLM training pipeline.
- Designed and implemented multi-turn agent interaction workflows, enabling agents to reason, plan, and adapt across sequential patient-agent interactions.

Research Assistant | University of Georgia

Aug 2024 - Now

(Agentic RL, Large Language Model)

- Researched agentic reinforcement learning for LLM-based agents, focusing on policy optimization over structured agent workflows.
- Modeled agent workflows as sequential decision processes, enabling long-horizon credit assignment across reasoning, planning, and execution stages.
- Designed reward signals and interaction protocols to guide agent behavior in multi-agent and single-agent settings.

Research Assistant | Columbia University

Jan 2021 - Apr 2021

(Natural Language Processing)

- Developed a Bi-LSTM model with an attention mechanism for sentiment analysis, achieving state-of-the-art performance with a 96% accuracy rate on the Amazon Musical Instruments Reviews dataset
- Innovated in natural language processing through the creation of an explainable AI model, enhancing interpretability by examining attention weight distributions